

CSE400 — Fundamentals of Probability in Computing

Lecture L11–12: Transformation of Random Variables

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1 Topic Title

Transformation of Random Variables and Functions of Two Random Variables

Topics covered:

- Transformation of random variables
- Function of two random variables
- Derived distribution example: $Z = X + Y$

2 Definitions and Notation

Let:

- X be a continuous random variable with known PDF $f_X(x)$
- $F_X(x)$ be the CDF of X

A transformation is defined as:

$$Y = g(X)$$

If invertible:

$$x = g^{-1}(y)$$

Goal:

$$f_Y(y) \quad \text{and} \quad F_Y(y)$$

For two random variables:

$$Z = g(X, Y)$$

Examples:

$$Z = X + Y, \quad X - Y, \quad \frac{X}{Y}, \quad XY$$

3 Assumptions / Conditions

3.1 Monotonic Transformation

The function $g(x)$ must be:

- monotonically increasing, or
- monotonically decreasing

4 Main Results / Theorems

4.1 Transformation Theorem (Single Random Variable)

Step 1: CDF Transformation

$$\begin{aligned} F_Y(y) &= P(Y \leq y) \\ &= P(g(X) \leq y) \end{aligned}$$

If g is increasing:

$$F_Y(y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$$

If g is decreasing:

$$F_Y(y) = P(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y))$$

Step 2: PDF Transformation

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right|$$

where

$$x = g^{-1}(y)$$

5 Proofs / Derivations

5.1 Function of Two Random Variables

Let

$$Z = X + Y$$

$$\begin{aligned} F_Z(z) &= P(Z \leq z) \\ &= P(X + Y \leq z) \end{aligned}$$

Integral Representation

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-y} f_{XY}(x, y) dx dy$$

5.2 Leibniz Rule

If

$$G(x) = \int_{a(x)}^{b(x)} h(x, y) dy$$

then

$$\frac{d}{dx} G(x) = \frac{db(x)}{dx} h(x, b(x)) - \frac{da(x)}{dx} h(x, a(x)) + \int_{a(x)}^{b(x)} \frac{\partial}{\partial x} h(x, y) dy$$

5.3 Differentiating the CDF

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

$$f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(z - y, y) dy$$

Equivalent form:

$$f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(x, z - x) dx$$

5.4 Independent Case (Convolution)

If X and Y are independent:

$$f_{XY}(x, y) = f_X(x)f_Y(y)$$

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x)f_Y(z - x) dx$$

This is called the **convolution integral**.

6 Worked Examples

6.1 Example 1 — Transformation of RV

Given:

$$X \sim \text{Uniform}(-1, 1)$$

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Find PDF of:

$$Y = \sin\left(\frac{\pi X}{2}\right)$$

Solution

$$x = \frac{2}{\pi} \sin^{-1}(y)$$

$$\frac{dx}{dy} = \frac{2}{\pi} \frac{1}{\sqrt{1 - y^2}}$$

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

$$f_Y(y) = \frac{1}{2} \cdot \frac{2}{\pi} \frac{1}{\sqrt{1 - y^2}}$$

$$f_Y(y) = \frac{1}{\pi \sqrt{1 - y^2}}, \quad -1 < y < 1$$

6.2 Example 2 — Sum of Independent Normals

$$X \sim N(0, 1), \quad Y \sim N(0, 1)$$

$$f_Z(z) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \cdot \frac{1}{\sqrt{2\pi}} e^{-(z-x)^2/2} dx$$

Result:

$$Z \sim N(0, 2)$$

6.3 Example 3 — Exponential RVs

$$X, Y \sim \text{Exponential}(\lambda)$$

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$

$$f_Y(y) = \lambda e^{-\lambda y}, \quad y > 0$$

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} dx$$

$$= \lambda^2 e^{-\lambda z} \int_0^z dx$$

$$f_Z(z) = \lambda^2 z e^{-\lambda z}, \quad z > 0$$

6.4 Example 4 — Difference of RVs

$$Z = X - Y$$

$$F_Z(z) = P(X - Y \leq z)$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{z+y} f_{XY}(x, y) dx dy$$

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

7 Additional Transformations Mentioned

$$Z = \frac{X}{Y}, \quad R = \sqrt{X^2 + Y^2}$$